

MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
UNIVERSITY OF CAMBRIDGE LOCAL EXAMINATIONS SYNDICATE
General Certificate of Education Advanced Level
Higher 2

MATHEMATICS

9758/02

Paper 2

October/November 2019

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** the questions.

Write your answers in the spaces provided in the Question Paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

1 You are given that $I = \int x(1-x)^{\frac{1}{2}} dx$.

(i) Use integration by parts to find an expression for I . [2]

(ii) Use the substitution $u^2 = 1 - x$ to find another expression for I . [2]

(iii) Show algebraically that your answers to parts (i) and (ii) differ by a constant.

[2]

- 2 (i) Sketch the graph of $y = \frac{2-x}{3x^2+5x-8}$. Give the equations of the asymptotes and the coordinates of the point(s) where the curve crosses either axis. [4]

(ii) Solve the inequality $\frac{2-x}{3x^2+5x-8} > 0$.

[1]

(iii) Hence solve the inequality $\frac{x-2}{3x^2+5x-8} > 0$.

[1]

- 3 A solid cylinder has radius r cm, height h cm and total surface area 900 cm^2 . Find the exact value of the maximum possible volume of the cylinder. Find also the ratio $r : h$ that gives this maximum volume. [7]

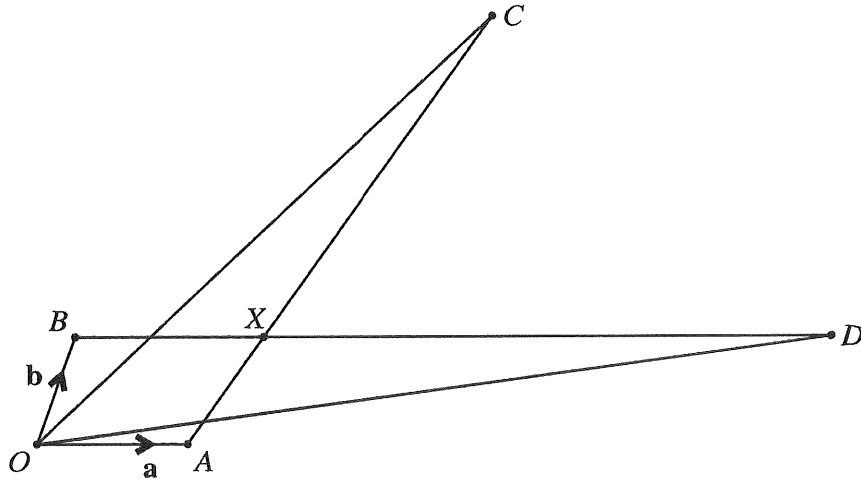
- 4 (i) Given that $f(x) = \sec 2x$, find $f'(x)$ and $f''(x)$. Hence, or otherwise, find the Maclaurin series for $f(x)$, up to and including the term in x^2 . [5]

(ii) Use your series from part (i) to estimate $\int_0^{0.02} \sec 2x \, dx$, correct to 5 decimal places. [2]

(iii) Use your calculator to find $\int_0^{0.02} \sec 2x \, dx$, correct to 5 decimal places. [1]

(iv) Comparing your answers to parts (ii) and (iii), and with reference to the value of x , comment on the accuracy of your approximations. [2]

(v) Explain why a Maclaurin series for $g(x) = \operatorname{cosec} 2x$ cannot be found. [1]



With reference to the origin O , the points A , B , C and D are such that $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$, $\vec{OC} = 2\mathbf{a} + 4\mathbf{b}$ and $\vec{OD} = \mathbf{b} + 5\mathbf{a}$. The lines BD and AC cross at X (see diagram).

- (i) Express $\vec{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

[4]

The point Y lies on CD and is such that the points O , X and Y are collinear.

(ii) Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio $OX : OY$.

[6]

Section B: Probability and Statistics [60 marks]

- 6** In a certain country there are 100 professional football clubs, arranged in 4 divisions. There are 22 clubs in Division One, 24 in Division Two, 26 in Division Three and 28 in Division Four.
- (i) Alice wishes to find out about approaches to training by clubs in Division One, so she sends a questionnaire to the 22 clubs in Division One. Explain whether these 22 clubs form a sample or a population. [1]
- (ii) Dilip wishes to investigate the facilities for supporters at the football clubs, but does not want to obtain the detailed information necessary from all 100 clubs. Explain how he should carry out his investigation, and why he should do the investigation in this way. [2]

- (iii) Find the number of different possible samples of 20 football clubs, with 5 clubs chosen from each division. [3]

- 7 A company produces drinking mugs. It is known that, on average, 8% of the mugs are faulty. Each day the quality manager collects 50 of the mugs at random and checks them; the number of faulty mugs found is the random variable F .
- (i) State, in the context of the question, two assumptions needed to model F by a binomial distribution. [2]

You are now given that F can be modelled by a binomial distribution.

- (ii) Find the probability that, on a randomly chosen day, at least 7 faulty mugs are found. [2]
- (iii) The number of faulty mugs produced each day is independent of other days. Find the probability that, in a randomly chosen working week of 5 days, at least 7 faulty mugs are found on no more than 2 days. [2]

The company also makes saucers. The number of faulty saucers also follows a binomial distribution. The probability that a saucer is faulty is p . Faults on saucers are independent of faults on mugs.

- (iv) Write down an expression in terms of p for the probability that, in a random sample of 10 saucers, exactly 2 are faulty. [1]

The mugs and saucers are sold in sets of 2 randomly chosen mugs and 2 randomly chosen saucers. The probability that a set contains at most 1 faulty item is 0.97.

- (v) Write down an equation satisfied by p . Hence find the value of p . [4]

- 8 Gerri collects characters given away in packets of breakfast cereal. There are four different characters: Horse, Rider, Dog and Bird. Each character is made in four different colours: Orange, Yellow, Green and White. Gerri has collected 56 items; the numbers of each character and colour are shown in the table.

	Orange	Yellow	Green	White
Horse	1	1	3	4
Rider	1	1	7	5
Dog	3	7	1	6
Bird	4	5	6	1

- (i) Gerri puts all the items in a bag and chooses one item at random.

- (a) Find the probability that this item is either a Horse or a Rider.

[1]

- (b) Find the probability that this item is either a Dog or a Bird but the item is not White.

[1]

- (ii) Gerri now puts the item back in the bag and chooses two items at random.

- (a) Find the probability that both of the items are Horses, but neither of the items is Orange.

[1]

- (b) Find the probability that Gerri's two items include exactly one Dog and exactly one item that is Yellow. [3]

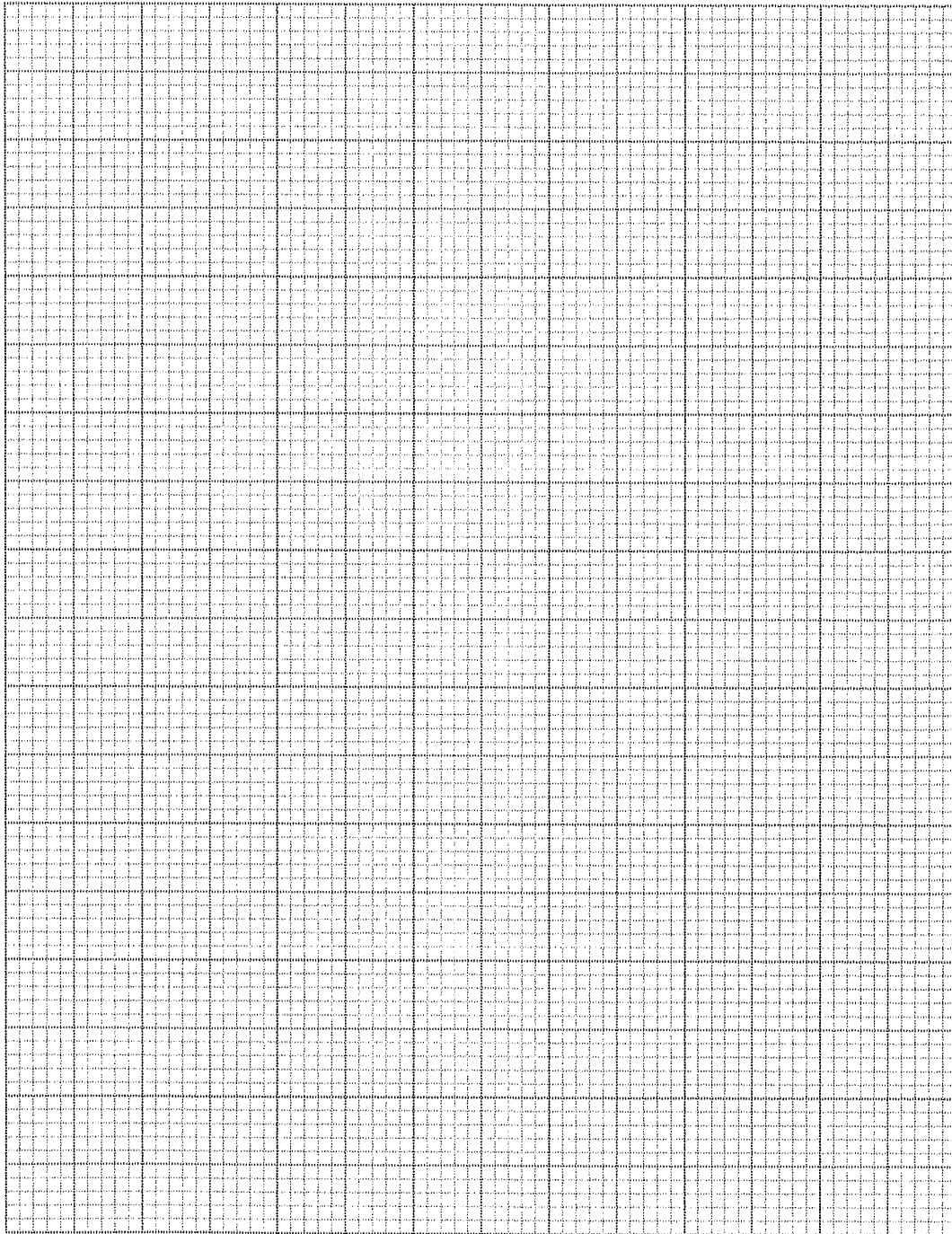
- (iii) Gerri has two favourites among the 16 possible colour/character combinations. The probability of choosing these two at random from the 56 items is $\frac{1}{77}$. Write down all the possibilities for Gerri's two favourite colour/character combinations. [3]

- 9 A company produces resistors rated at 750 ohms for use in electronic circuits. The production manager wishes to test whether the mean resistance of these resistors is in fact 750 ohms. He knows that the resistances are normally distributed with variance 100 ohms².
- (i) Explain whether the manager should carry out a 1-tail test or a 2-tail test. State hypotheses for the test, defining any symbols you use. [2]

The production manager takes a random sample of 8 of these resistors. He finds that the resistances, in ohms, are as follows.

742 771 768 738 769 752 742 766

- (ii) Find the mean of the sample of 8 resistors. Carry out the test, at the 5% level of significance, for the production manager. Give your conclusion in context. [5]



10 [Continued]

Bhani models the same data using a straight line passing through the points (40, 22) and (55, 17). The sum of the squares of the residuals for Bhani's line is 1.

(ii) State, with a reason, which of the two models, Abi's or Bhani's, gives a better fit. [1]

(iii) State the coordinates of the point that the least squares regression line must pass through. [1]

(iv) Use your calculator to find the equation of the least squares regression line of y on x . State the value of the product moment correlation coefficient. [3]

- (v) Use the equation of the regression line to estimate the fuel consumption when the speed is 30 kilometres per hour. Explain whether you would expect this value to be reliable. [2]

- (vi) Cerie performs a similar experiment on a different car. She finds that the sum of the squares of the residuals for her line is 0. What can you deduce about the data points in Cerie's experiment? [1]

- 11 In this question you should state clearly all the distributions that you use, together with the values of the appropriate parameters.**

Arif is making models of hydrocarbon molecules. Hydrocarbons are chemical compounds made from carbon atoms and hydrogen atoms. Arif has a bag containing a large number of white balls to represent the carbon atoms, and a bag containing a large number of black balls to represent the hydrogen atoms. The masses of the white balls have the distribution $N(110, 4^2)$ and the masses of the black balls have the distribution $N(55, 2^2)$. The units for mass are grams.

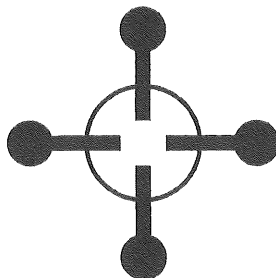
- (i) Find the probability that the total mass of 4 randomly chosen white balls is more than 425 grams. [2]

- (ii) Find the probability that the total mass of a randomly chosen white ball and a randomly chosen black ball is between 161 and 175 grams. [2]

- (iii) The probability that 2 randomly chosen white balls and 3 randomly chosen black balls have total mass less than M grams is 0.271. Find the value of M . [4]

11 [Continued]

Arif also has a bag containing a large number of connecting rods to fix the balls together. The masses of the connecting rods, in grams, have the distribution $N(20, 0.9^2)$. In order to make models of methane (a hydrocarbon), Arif has to drill 1 hole in each black ball, and 4 holes in each white ball, for the connecting rods to fit in. This reduces the mass of each black ball by 10% and reduces the mass of each white ball by 30%.



A methane molecule consists of 1 carbon atom and 4 hydrogen atoms. Arif makes a model of a methane molecule using 4 black balls, 1 white ball and 4 connecting rods (see diagram). The balls and connecting rods are all chosen at random.

(iv) Find the probability that the mass of Arif's model is more than 350 grams.

[4]

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